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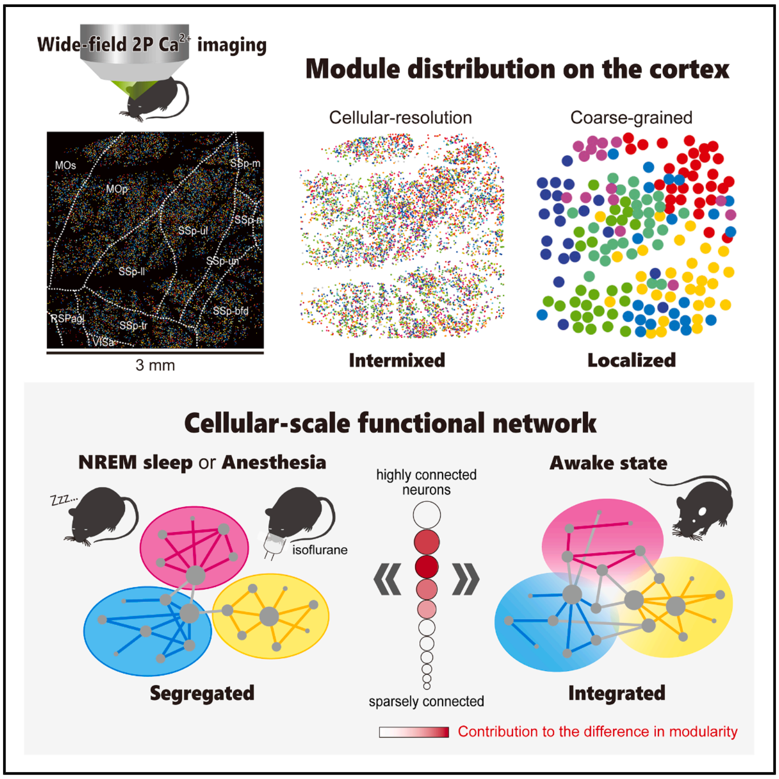
Single-cell resolution functional networks during unconsciousness are segregated into spatially intermixed modules

Graphical abstract



Highlights

- Cellular-resolution wide-field calcium imaging enables multi-scale network analysis
- Cellular-scale networks in sleep and anesthesia are more segregated than wakefulness
- Segregation is apparent only at the cellular level but not at the mesoscale
- Modules are intermixed at the cellular scale but localized at the mesoscale

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In brief

By recording cellular-resolution neural activity across multiple cortical regions, Kiyooka, Oomoto et al. show that brain networks become more segregated into modules during sleep and anesthesia, driven by moderately connected neurons. Despite this functional separation, these modules are not spatially clustered but remain physically intermixed, indicating a complex organizational principle.

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Single-cell resolution functional networks during unconsciousness are segregated into spatially intermixed modules

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SUMMARY

The common neural mechanisms underlying the reduction of consciousness during sleep and anesthesia remain unclear. Previous studies have examined changes in network structure by only using recordings with limited spatial resolution, which has hindered the investigation of the critical spatial scales for the reduction of consciousness. To address this issue, we recorded calcium signals from approximately 10,000 neurons across multiple cortical regions in awake, sleeping, and anesthetized mice and compared network structure at different spatial scales by leveraging single-cell resolution and wide-field two-photon microscopy. At the single-cell scale, both sleep and anesthesia exhibit higher network modularity than an awake state, indicating a segregated network, but modules are spatially intermixed in all three states. In contrast, at the mesoscale, there are no consistent differences in modularity between states, and modules are spatially localized. Our multi-scale analysis challenges the traditional view of network segregation during unconsciousness and indicates a scale-dependent network organization.

INTRODUCTION

The process by which consciousness diminishes during sleep and anesthesia remains elusive in neuroscience. Prevailing theories of consciousness propose that the brain's ability to integrate information is an indispensable prerequisite for consciousness, given the unified nature of the conscious experience.[1–4] Such information integration is enabled by the networked interactions of neurons at the microscale and brain regions at the macro/mesoscale. Therefore, it is crucial to examine to what extent and how the brain network is integrated or segregated at both the micro- and macro/mesoscale depending on the brain state.[5–7] Such network-level understanding is pivotal for elucidating the possibly shared neural underpinnings of reduced consciousness during sleep and anesthesia.

Current empirical findings from studies using macro/mesoscale neural recordings, such as functional MRI (fMRI) and electroencephalogram (EEG), suggest that functional networks are more segregated during non-rapid eye movement (NREM) sleep and anesthesia than during an awake state[8–18]; that is, interactions across brain regions are diminished. For example, a perturbational approach using transcranial magnetic stimulation (TMS) has shown that TMS-evoked EEG potentials do not propagate across brain regions during NREM sleep but do so during an awake state, implying a breakdown of cortical effective connectivity during NREM sleep.[13] Another approach, using a graph-theoretic measure called modularity, has shown that functional networks are more segregated into spatially localized modules during NREM sleep[10,17] as well as during propofol-induced[19] and isoflurane-induced anesthesia[9,16] than during an awake state.

While many past studies have investigated macro/mesoscale properties of networks,[9,10,12,16–18,20–31] where the scales are mainly determined by the constraints of measurement technique, crucial spatial scales for understanding the mechanisms underlying the reduction of consciousness remain to be elucidated.[4,32–34] To address this, it is essential to record neural activity at the single-cell resolution for two reasons. First, neurons represent one of the most fundamental units for information processing in the brain.[35] Second, by spatially coarse-graining single-cell resolution recordings, we can investigate macro/mesoscale properties of the network with signals directly derived from neural activity. The resulting macro/mesoscale data are qualitatively distinct from those obtained through conventional recording techniques, such as fMRI and EEG, which measure indirect proxies of neural activity. Although a few studies have investigated single-cell resolution functional networks, these studies have analyzed data from a limited brain area[36–40] or a limited number of neurons[41–43] because of the narrow spatial coverage of recording sites. These constraints make it difficult to investigate coarser spatial scales and cross-regional interactions, both of which are crucial for understanding brain network structures associated with the reduction of consciousness. Consequently, little is known about cellular-resolution functional networks across multi-modal brain regions and their relationship to networks at coarser spatial scales.

To uncover the changes in functional network structures commonly associated with the reduction of consciousness from the single-cell scale to the mesoscale, we recorded the activity of approximately 10,000 neurons during an awake state, sleep, and isoflurane-induced anesthesia using fast, single-cell resolution, and wide field of view (FOV) two-photon laser scanning microscopy, FASHIO-2PM (fast-scanning high optical invariant two-photon microscopy), recently developed by our group.[44,45] The FOV spanning a contiguous 3×3 mm area encompasses multiple cortical regions, including the somatosensory cortex, motor cortex, and other cortices, which enables us to explore the cellular-resolution functional networks over expansive brain areas. Importantly, by recording neural activity during sleep for the first time at single-cell resolution across multi-modal brain regions, we can investigate changes in network structure shared between sleep and anesthesia compared to an awake state, revealing common structures associated with reduced consciousness.

To investigate to what extent and how the networks are segregated, we compare network modularity[46,47] and the spatial distribution of modules during NREM sleep and anesthesia with those during an awake state, both at the single-cell and mesoscale levels. We show that modularity of single-cell resolution functional networks is higher during NREM sleep and anesthesia than during an awake state, with differences associated with mid- to high-degree neurons rather than the highest-degree neurons. However, we show that these modules are distributed in a spatially intermixed manner in all states. On the other hand, modularity of mesoscale functional networks estimated from coarse-grained neural activity is not consistently different between states, and the identified modules are spatially localized. Our results provide insights into the cellular-scale organization of functional networks universally associated with reduced consciousness and highlight a scale-dependent organization of network structures.

RESULTS

Framework for investigating network structure

We investigate the structure of functional networks during an awake state, NREM sleep, and isoflurane-induced anesthesia, particularly in terms of network segregation. Modularity is used as a measure to quantify the degree of network segregation. It quantifies the extent to which a network can be divided into non-overlapping "modules," where neurons within the same modules are densely connected, but those with different modules are sparsely connected. The lower the modularity, the more integrated the network, whereas the higher the modularity, the more segregated the network (Figure 1A)[48] (STAR Methods).

Modularity is central to this study because it allows us to investigate not only the extent of segregation but also the manner in which networks are segregated or integrated. Calculating modularity involves identifying module assignments so that neurons within the same module have dense connections, while neurons in different modules have sparser connections. This approach gives us two key insights into how networks are segregated or integrated. First, we can characterize neurons based on the contribution to the modular structure of the networks. Neurons that have many across-module connections but few within-module connections are referred to as connector nodes and contribute to network integration, while neurons that have few across-module connections but many within-module connections are referred to as provincial nodes and contribute to network segregation (Figure 1B). Analyzing these neurons provides a detailed single-cell perspective on how network segregation occurs. Second, we can examine the spatial distribution of the identified modules and gain insight into how functional networks are organized across brain regions.

Fast, cell-resolution, contiguous, wide two-photon calcium imaging across multi-modal cortical areas during an awake state, sleep, and anesthesia

We recorded the cortex-wide activity of individual neurons in waking, sleeping, and anesthetized mice (Figure 2), using FASHIO-2PM.[44,45] The FOV spanning a contiguous 3×3 mm area allowed us to record multiple brain regions in cortical layer 2/3 excitatory neurons, including the somatosensory, motor, and higher-order cortices (Figure 2; Video S1). To determine the sleep stages in mice, we simultaneously recorded electromyogram (EMG) potentials from neck muscles and EEG signals from screw electrodes placed on the lateral secondary visual area (V2L) of the imaged hemisphere (Figure 2B) (STAR Methods). Four brain states were determined by post hoc analysis of the EEG and EMG: an awake state, NREM sleep, rapid eye movement (REM) sleep, and a quiet awake state. Although REM sleep and a quiet awake state were occasionally observed, these states were rare and we could not record data long enough to reliably estimate correlation during these states (see Table S1 for the duration of these states). Therefore, we focused on an awake state and NREM sleep for subsequent analysis. To record neural activity during anesthesia, we anesthetized mice with 0.6% isoflurane inhalation after awake recording (hereafter, we use the terms "sleep" and "anesthesia" for NREM sleep and isoflurane-induced anesthesia, respectively, in the main text unless otherwise specified).

To prepare for network analysis, we investigated how the number of active neurons differed between brain states. This is because our network analysis focuses on neurons that have at least one Ca^{2+} event within all windows used to estimate the networks in each session, ensuring that a network size within a session is the same. Here, a session refers to one continuous recording comprising either a wakefulness-sleep or a wakefulness-anesthesia time series (each column in Tables S1 and S2

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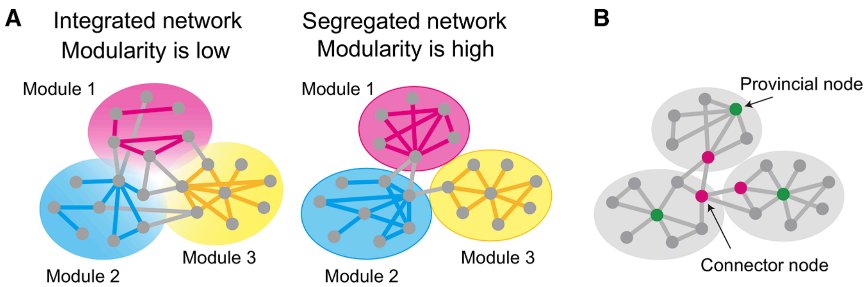

Figure 1. Schematics of the concept of modularity and two types of neurons

(A) Integrated network with low modularity (left) and segregated network with high modularity (right). In this study, a node corresponds to a single neuron or a group of neurons, and an edge represents a highly correlated node pair. Edges within each module are colored according to the respective module, and edges between modules are colored gray.

(B) Connector nodes (magenta) and provincial nodes (green) in a network. Connector nodes have few within-module but many across-module connections, while provincial nodes have many within-module but few across-module connections.

corresponds to a single session). We found that the number of neurons that have at least one Ca^{2+} event within each time-window was, on average, 65% during anesthesia due to the reduction of neural activity caused by isoflurane inhalation, while it was above 95% during an awake state and sleep (Figure S1), although the reduction of neural activity was also observed during sleep (see Figures S2 and S3 for event frequency and mean amplitude of all mice, respectively). This led to a substantial reduction of the number of neurons used in the following network comparisons between an awake state and anesthesia, from approximately 4,000–7,000 to 2,000–3,000 (see Tables S1 and S2 for the number of neurons used for network analysis). Although neural activity is reduced during anesthesia compared to an awake state, it is not evident whether the network structure of active neurons differs between these states, which is investigated in the following sections.

Segregated functional networks during sleep and anesthesia

In this study, we separately compared functional networks during an awake state with those during sleep and with those during anesthesia. This is because wakefulness-sleep comparison and wakefulness-anesthesia comparison are based on recordings from the same session, whereas sleep and anesthesia were recorded in separate sessions, making direct comparisons between these states challenging. As mentioned earlier, because the number of neurons active during anesthesia is significantly smaller compared to an awake state and sleep, the number of neurons available for comparing network structures between an awake state and anesthesia is much smaller than that for an awake state and sleep. In the following, we investigate whether there are commonalities between sleep and anesthesia from the perspective of changes in network modularity compared to an awake state.

To compare modularity (1) during an awake state and sleep and (2) during an awake state and anesthesia, we estimated functional networks in the three states. To do so, we first calculated Pearson correlation coefficients for all pairs of neurons based on their deconvolved Ca^{2+} signals. We found that the average Pearson correlation was higher during sleep and anesthesia than during an awake state (Figure DS1-1). While this global difference in connectivity strength is informative, our primary focus is on the pattern of connections rather than the variation of the overall connection strength. Therefore, to isolate the effect of connection patterns, we binarized the absolute correlation matrices at a threshold set so that the connection density was equal to a certain value, K[49] (STAR Methods) (Figures 3A and 3B; see also Figures S5 and S6 for the relationship between connection density and absolute correlation threshold during each state). To demonstrate the robustness of the results across a wide range of K, we varied K from 0.008 to 0.3, which we considered as the lowest and highest values, respectively, for valid network analysis. When K was 0.008, the threshold absolute correlation value in some mice exceeded 0.45, which could have led to the absence of edges in highly correlated neuron pairs. Thus, for $K \leqq 0.008$, we consider the networks unsuitable for graph-theoretic analysis. When K was 0.3, the threshold absolute correlation value in some mice was below 0.05, which can lead to the presence of edges between non-significantly correlated neuron pairs. Thus, for $K \geq 0.3$, we consider the networks unsuitable for graph-theoretic analysis.

To compute the modularity of the estimated functional networks, we first identified a module assignment that maximized the modularity index Q in Equation 3 using the Louvain algorithm. The adjacency matrices, with neuron IDs sorted according to the identified modules, are shown in the bottom rows of Figures 3A and 3B. Each block along the diagonal corresponds to a module. The number of detected modules was not significantly different between an awake state and sleep across different connection densities (Figure S7) (the 95% confidence intervals for the differences in the number of modules, shown as gray shaded areas in Figure S7C, include zero across connection densities). The number of detected modules was not consistently different across mice between an awake state and anesthesia when $0.03 \leq K$, but larger during anesthesia than during an awake state when $K < 0.03$ (Figure S8). After module assignment, we computed the modularity using Equation 3.

We found that modularity was higher during both sleep and anesthesia than during an awake state across different connection densities (Figures 3C–3F) (the 95% confidence intervals for the modularity differences, shown as gray shaded areas in Figures 3D and 3F, remained negative across connection densities). This suggests that functional networks during sleep and anesthesia are more segregated into modules than those during an awake state. This difference could not be explained by the number of modules because the number of modules was not significantly different between an awake state and sleep across all of the connection densities nor between an awake state and

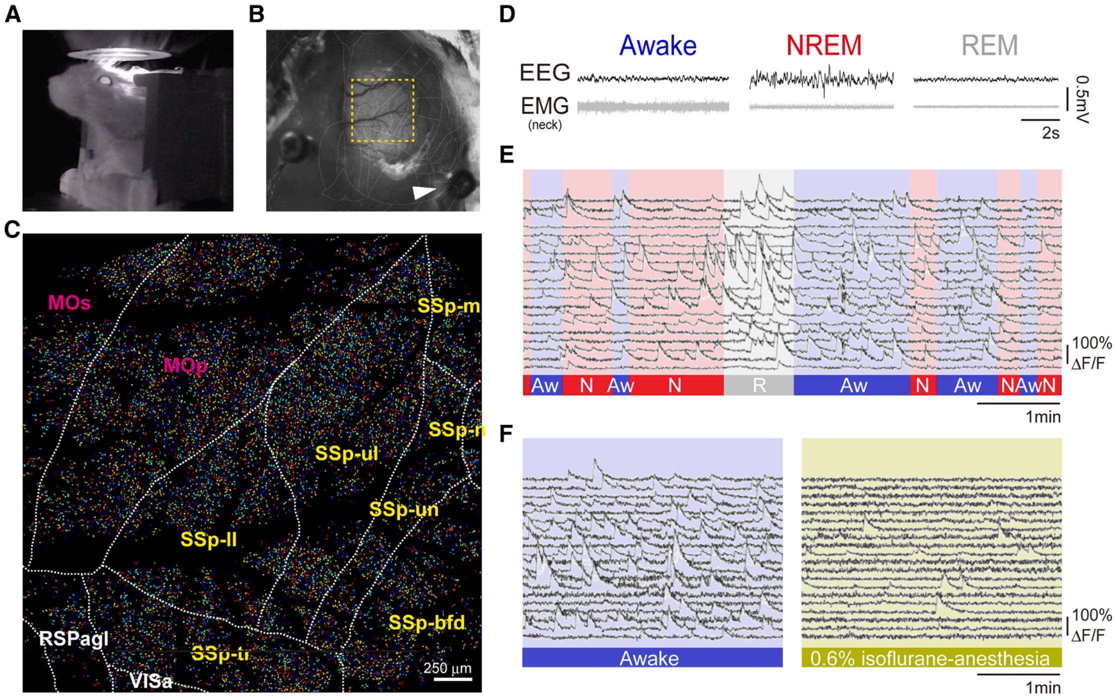

Figure 2. Fast, cell-resolution, wide-field two-photon calcium imaging from mice during an awake state and sleep

(A) An infrared camera image of a mouse under the FASHIO-2PM.

(B) A dorsal view of the mouse head. A yellow dashed square indicates the FOV. An arrowhead indicates a screw to acquire EEG from the same hemisphere on the imaging side. Overlaid white lines show borders according to the Allen Mouse Common Coordinate Framework.

(C) Activity of > *rbin* 10,000 neurons was recorded from a 3 × 3 mm FOV that included multiple brain regions. Borders between the regions are shown as white dashed lines according to the Allen Mouse Common Coordinate Framework. MOp, primary motor; MOs, secondary motor; SSp-bfd, S1-barrel field; SSp-ll, S1-lower limb, SSp-m, S1-mouse; SSp-n, S1-nose; SSp-tr, S1-trunk; SSp-ul, S1-upper limb; SSp-un, S1-unassigned area; RSPagl, retrosplenial area, lateral agranular part.

(D) Examples of EEG and EMG recordings during the imaging session. The sleep-wake state was estimated by post hoc analysis of EEG and EMG (STAR Methods).

(E) Representative Ca^{2+} signals from 20 randomly selected neurons in sleep-wake states. The bottom color bar indicates the sleep-wake state of the mouse: awake state (Aw, blue), NREM sleep (N, red), and REM sleep (R, gray).

(F) Representative Ca^{2+} signals from 20 randomly selected neurons in awake or anesthetic states.

anesthesia (Figures S7 and S8). Furthermore, even when we set the number of modules to be the same (STAR Methods), modularity remained higher during sleep and anesthesia than during an awake state, although, at $K = 0.15$ and 0.2, the variability of modularity differences between wakefulness and anesthesia became large, and the 95% confidence intervals included zero (Figures DS1-2 and DS1-3). Therefore, sleep and anesthesia are qualitatively similar in terms of the extent of network segregation compared with an awake state.

To confirm the robustness of the modularity results, we also recalculated modularity using three different resolution parameters (γ = 0.5, 1.5, 2.0) at a connection density $K = 0.05$ (STAR Methods) and obtained consistent results. We found that modularity was higher during both sleep and anesthesia than during an awake state (Figures DS1-4 and DS1-5). We note that, at $\gamma = 0.5$, the differences were not significant; this occurred because some networks resolved into a single module under this setting (Figures DS1-6 and DS1-7), for which a binary network's Q value is always 0.5.

To further verify the robustness of the modularity results, we investigated other measures of network integration or segregation. The most basic and widely used measures are global efficiency and transitivity (STAR Methods), which are related to small-world-ness and small-world propensity investigated in our previous study.[45] Global efficiency is defined as the average inverse shortest path length, quantifying how efficiently one node can communicate with another. Therefore, high global efficiency is associated with network integration. Transitivity is defined as the fraction of triangles in a network, quantifying the prevalence of local clustering. Therefore, high transitivity is associated with network segregation. We found that global efficiency was higher during an awake state than during sleep and anesthesia, when $K < 0.05$, but not different when $K \geq 0.05$ (Figures DS1-8 and DS1-9). We also found that transitivity was higher during sleep and anesthesia than during an awake state across different connection densities ($0.01 \leq K \leq 0.3$) (Figures DS1-10 and DS1-11). Although global efficiency could not clearly distinguish the brain states at higher densities, these results on global efficiency and transitivity are broadly consistent with increased modularity during sleep and anesthesia. These results confirm the robustness of the modularity result.

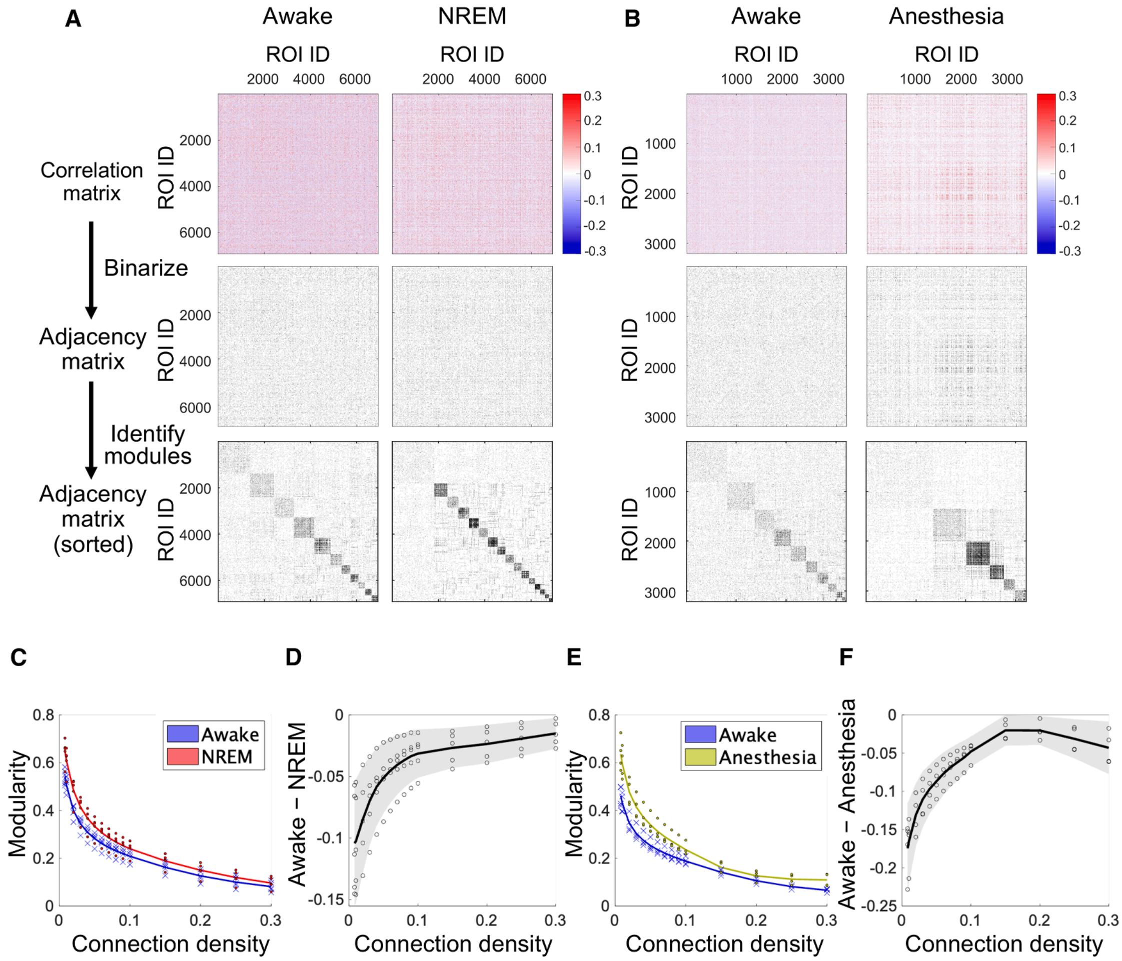

Figure 3. Estimation of functional networks and comparison of modularity

(A) Procedures for estimating functional networks and calculating modularity during an awake state (left) and NREM sleep (right).

(B) Procedures for estimating functional networks and calculating modularity during an awake state (left) and anesthesia (right).

(C) Comparison of modularity between an awake state (blue) and NREM sleep (red). Crosses (awake) and dots (NREM sleep) represent modularity of individual mice, averaged across networks estimated from different time windows. Lines represent the mean modularity across mice.

(D) Difference in modularity between an awake state and NREM sleep. Each circle represents an individual mouse. The black line represents the average across mice. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample *t* test, $n = 5$, $df = 4$).

(E) Same as (C) but comparing an awake state (blue) and anesthesia (yellow).

(F) Same as (D) but comparing an awake state and anesthesia. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample *t* test, $n = 4$, $df = 3$).

Highest-degree neurons do not necessarily contribute to differences in modularity

Although we found in the previous section that sleep and anesthesia exhibited similar overall network trends in terms of increased modularity, it remains unclear whether modularity differences result from the influence of a small number of neurons or from the broader population of neurons. Such an analysis is helpful for understanding whether segregation patterns during sleep and anesthesia are similar at the single-cell level. Specifically, we focused on a neuron's node degree, as it is a fundamental characteristic of network nodes, and high-degree neurons can have a large influence on network structure. To determine whether modularity differences are associated with a small number of highest-degree neurons or broader populations, our analysis proceeds in three steps. First, we propose a measure to quantify each neuron's contribution to modularity. Second, we confirm that our proposed measure captures such a contribution by comparing the histogram of it across states. Third, we investigate their relationship with node degree.

To investigate each neuron's contribution to modularity, we propose a node-specific modularity measure, Q_i, by decomposing the overall modularity, Q, into contributions of each node:

$$Q = \sum_{i \in G} Q_i, \qquad \text{(Equation 1)}$$

where G is the set of all nodes in the network. This measure quantifies the extent to which a node i contributes

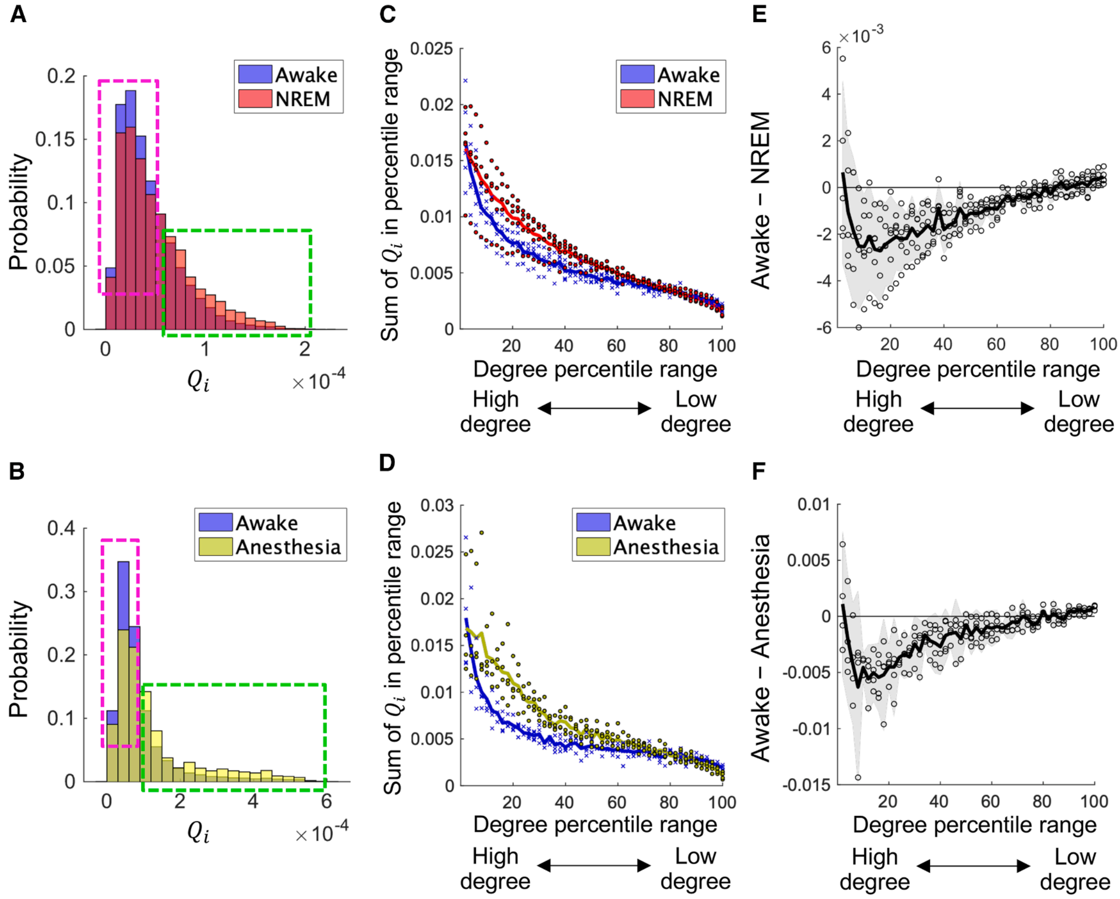

Figure 4. Contribution of each neuron to modularity at $K = 0.05$

For the data of different K, see Figures DS1-12 and DS1-13.

(A) Representative histogram of Q_i during an awake state (blue) and NREM sleep (red). See also Figures DS1-12 for the histograms of all mice. The green dashed line encloses neurons with high Q_i (provincial-like neurons), and the magenta dashed line encloses those with low Q_i (connector-like neurons).

(B) Representative histogram of Q_i during an awake state (blue) and anesthesia (yellow). See also Figure DS1-12 for the histograms of all mice.

(C) Comparison of contributions to modularity across degree percentiles between an awake state (blue) and NREM sleep (red). For a given percentile range (x axis value x), the sum of Q_i across neurons ranked in the top x–2% to x% by degree are plotted. Crosses (an awake state) and dots (NREM sleep) represent these percentile-based sums of Q_i for individual mice, averaged across networks estimated from different time windows. Lines represent the mean value across mice.

(D) Same as (C) but comparing an awake state (blue) and anesthesia (yellow).

(E) Difference in the percentile-based sum of modularity between an awake state and NREM sleep. Each circle represents an individual mouse. The black line represents the average across mice. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample t test, $n = 5$, $df = 4$).

(F) Same as (E) but comparing an awake state and anesthesia. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample t test, $n = 4$, $df = 3$).

to the modularity Q (see STAR Methods for details) and is defined as

$$Q_i = \frac{1}{2l}\left[\frac{\sum_{j \notin G_i} k_j}{2l} k_i^{\text{within}} - \frac{\sum_{j \in G_i} k_j}{2l} k_i^{\text{between}}\right], \qquad \text{(Equation 2)}$$

where l is the number of edges in the network, G_i is the set of nodes that belong to the same module as node i, k_j is the degree of node j, k_i^{within} is the within-module degree of a node i, and k_i^{between} is the across-module degree of a node i. Intuitively, the higher the within-module degree k_i^{within} and the lower the across-module degree k_i^{between}, the higher the Q_i, which indicates a greater contribution to modularity. In other words, neurons with high Q_i play a provincial node-like role, whereas neurons with low Q_i play a connector node-like role (Figure 1B).

To confirm that our proposed Q_i measure captures each neuron's contribution to modularity, we compared the histogram of Q_i between an awake state and sleep as well as between an awake state and anesthesia. As expected, we found that the number of neurons that exhibited high Q_i values (neurons enclosed by the green dashed line in Figure 4A) was larger during sleep than during an awake state, while the number of neurons that exhibited low Q_i values (neurons enclosed by the magenta dashed line in Figure 4A) was smaller during sleep than during an awake state (see Figure 4A for representative data; see Figures DS1-12A–DS1-12C for other connection densities and other mice). Similarly, the number of neurons that exhibited high Q_i values (neurons enclosed by the green dashed line in Figure 4B) was larger during anesthesia than during an awake

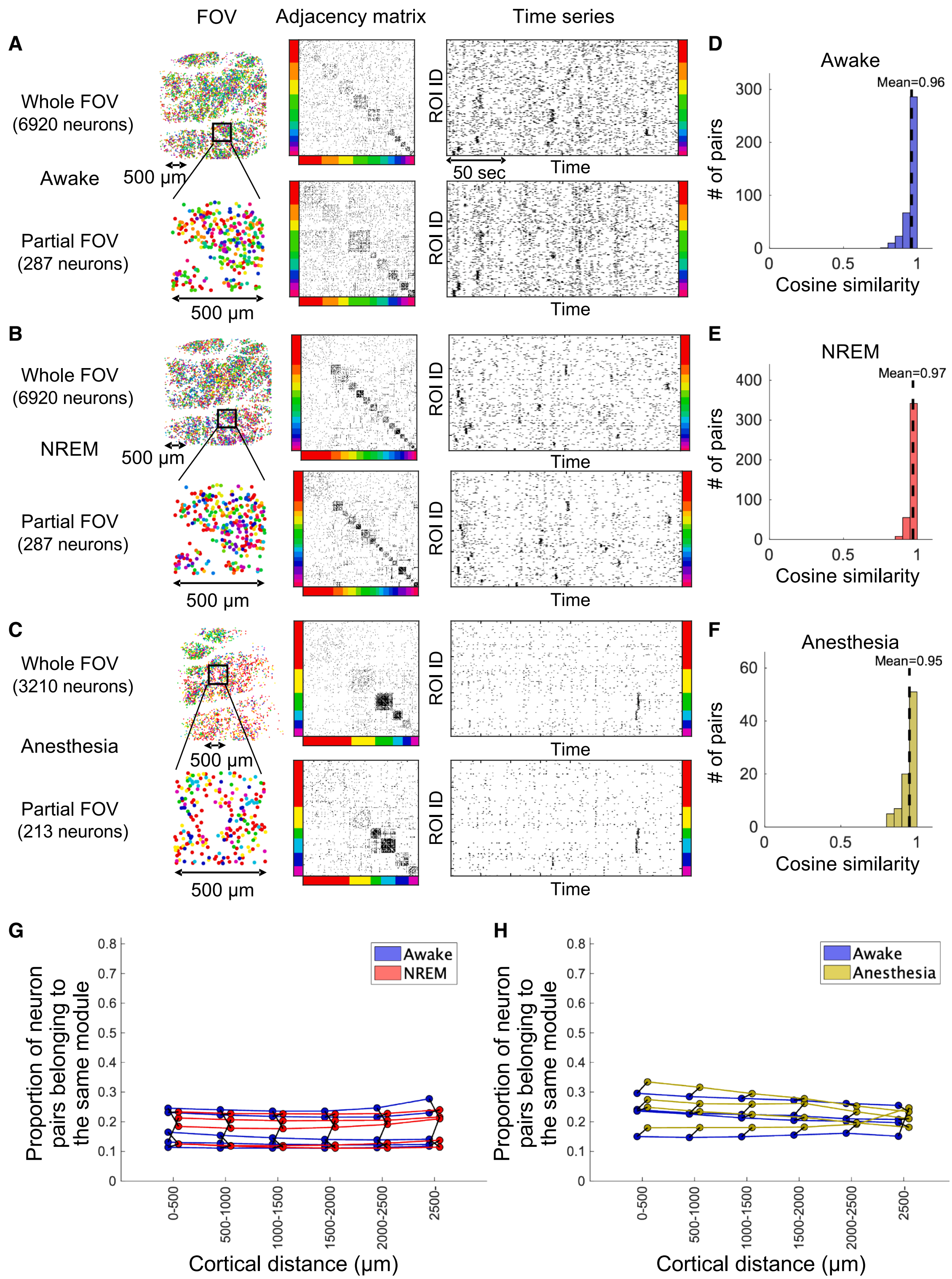

Figure 5. Spatially intermixed distribution of single-cell resolution functional network modules at K = 0.05

(A) Spatial distribution of modules (left), adjacency matrix (middle), and time series (right) for both the entire FOV (top) and the partial FOV (bottom) of representative functional networks during an awake state. Neurons within the adjacency matrices and time series are sorted by their respective modules. Each color represents a specific module.

(B) Same as (A) but during NREM sleep.

(legend continued on next page)

state, while the number of neurons that exhibited low Q_i values (neurons enclosed by the magenta dashed line in Figure 4B) was smaller during anesthesia than during an awake state (see Figure 4B for representative data; see Figures DS1-12D–DS1-12F for other connection densities and other mice). These results suggest that functional networks during both sleep and anesthesia exhibit similar segregation patterns, characterized by a decrease in the number of connector-like neurons and an increase in the number of provincial-like neurons.

Having confirmed how Q_i captures each neuron's contribution to modularity, we examined the relationship between Q_i and the degree. The conventional view holds that hub neurons—those with the highest degrees—play key roles in shaping network structure and computation.[50–52] One might therefore hypothesize that these neurons would contribute substantially to state-dependent differences in modularity. We found, however, that neurons with degrees in the top 10%–60% percent, rather than the top 10%, were primarily responsible for the differences in modularity between an awake state and sleep when $K = 0.05$ (Figures 4C and 4E) (Figure 4E shows that the 95% confidence interval for the difference in the contribution to modularity, based on two-tailed one-sample t test, $n = 5$, $df = 4$, is negative within the 10%–60% percent range). Likewise, neurons with degrees in the top 10%–40% percent, rather than the top 10% percent, were primarily responsible for the differences in modularity between an awake state and anesthesia when $K = 0.05$ (Figures 4D and 4F) (Figure 4F shows that the 95% confidence interval for the difference in the contribution to modularity, based on two-tailed one-sample t test, $n = 4$, $df = 3$, is negative within the 10%–60% percent range). Although the specific percentage ranges that contribute to modularity differences varied depending on K, the highest-degree neurons were not consistently related to modularity differences across different K values (Figure DS1-13). Nonetheless, these highest-degree neurons made a relatively large contribution to the total network modularity compared with other degree ranges. These results suggest that sleep and anesthesia exhibit a similar pattern in which mid- to high-degree neurons, rather than the highest-degree neurons, are primarily associated with the observed differences in modularity relative to an awake state.

We next quantitatively confirmed this result in two ways. First, we calculated the cosine similarity of the module-proportion vectors between the partial FOVs and the entire FOV. For the wakefulness-sleep data, we used non-overlapping 0.5×0.5 mm regions with more than 250 neurons as partial FOVs. For the wakefulness-anesthesia data, due to the smaller number of analyzed neurons, we used those with more than 100 neurons as partial FOVs. For each partial FOV, we constructed a vector whose elements are the number of neurons belonging to each module and then computed the cosine similarity between that vector and its counterpart for the entire FOV. The cosine similarity was, on average, higher than 0.95 during all states ($K = 0.05$) (Figures 5D–5F; see Figures DS1-14A and DS1-14C for different K). This indicates that the module composition within the most partial FOVs is similar to that of the entire FOV. Second, we examined the relationships between cortical distance and the probability of two neurons belonging to the same module. We found that the probability did not depend on cortical distance (see Figures 5G and 5H for $K = 0.05$; see Figures DS1-14B and DS1-14D for different K). This suggests that nearby neurons do not necessarily belong to the same module, while distant neurons sometimes do. These two analyses quantitatively confirm that modules are distributed in a spatially intermixed manner.

We confirmed that these results did not depend on the choice of module detection method by performing complementary analyses from two perspectives. First, we investigated modules partitioned by consensus clustering. In the main figures, we show data for the partitions that achieve the maximal Q value among 200 Louvain iterations (Figures 3, 4, 5, 6, and 7). Although this strategy is effective for identifying the most modular partition and comparing its Q across networks, these partitions are not necessarily the most representative. As network size increases, the number of partitions that achieves near-maximal Q grows exponentially, and partitions with similar Q values can differ substantially.[53] A conventional solution is consensus clustering, which yields representative partitions across iterations[54,55] (STAR Methods). Second, we examined modules detected with different resolution parameters (STAR Methods). In both cases, our main findings held regardless of whether we used max-Q or consensus clustering across different values of γ (Figures DS1-15, DS1-16, and DS1-17).

Spatially intermixed modules of single-cell resolution functional networks

To examine whether modules are spatially localized, as observed in macro/mesoscale functional networks,[17,23,30,56] or intermixed on the cortex, we examined the spatial distribution of modules. We found that the modules were spatially intermixed during an awake state (Figure 5A), sleep (Figure 5B), and anesthesia (Figure 5C). This indicates that adjacent neurons do not necessarily belong to the same module, while distant neurons sometimes belong to the same module (left column in Figures 5A–5C), although there is a little spatial imbalance of the distribution of modules. Thus, even during sleep and anesthesia, when functional networks are "segregated," long-range functional connections remain across brain regions. This spatial distribution was consistent across all states, although the degree

(C) Same as (A) but during anesthesia.

(D–F) Histogram of cosine similarity of module composition between the entire FOV and partial FOVs during an awake state (D), NREM sleep (E), and anesthesia (F). Each histogram shows data of all windows across all mice. Dotted lines represent the mean cosine similarity. For the data of other K, see Figures DS1-14A and DS1-14C.

(G and H) Relationships between cortical distance and module assignments of single-cell resolution functional networks ($K = 0.05$) during an awake state (blue), NREM sleep (red), and anesthesia (yellow). The x axis represents the cortical distance, and the y axis represents the proportion of neuron pairs belonging to the same modules. Each dot represents an individual mouse, with data averaged across networks estimated from different time windows. Data from the same mouse are connected with lines. For the data of other K, see Figures DS1-14B and DS1-14D.

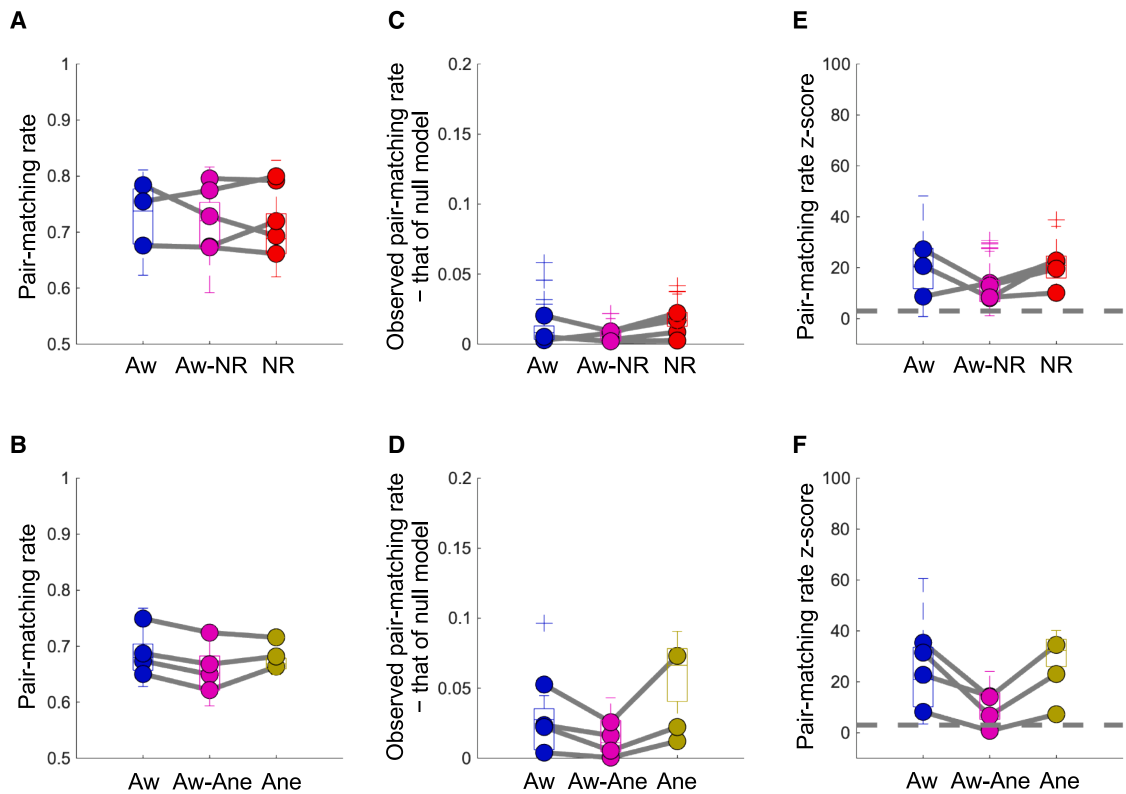

Figure 6. Temporal stability of modules

(A and B) The pair-matching rate represents the proportion of neuron pairs consistently assigned to the same module across two windows. "Aw," "NR," and "Ane" represent the stability of modules within an awake state, NREM sleep, and anesthesia, respectively. "Aw-NR" and "Aw-Ane" represent the stability of modules between an awake state and NREM sleep and between an awake state and anesthesia, respectively. Each dot corresponds to an individual mouse, with the value averaged across networks estimated from different time windows. In within-state comparisons, data without dots correspond to individuals where only one time window was available. The boxplots show data of all windows without distinguishing individuals, and crosses represent outliers.

(C and D) Observed pair-matching rate minus the null model pair-matching rate.

(E and F) The Z score of (A) and (B). The gray dashed lines represent a Z score value of 3.

of segregation differed between an awake state and sleep as well as between an awake state and anesthesia.

Temporal stability of modules

To investigate whether spatially intermixed modules identified in the previous section are randomly reconfigured or possess a temporally stable structure, we calculated the similarity of module assignments between different time windows. Initially, we computed the proportion of neuron pairs that were consistently assigned to the same module across two windows. We found that the observed matching rates ranged from 60% to 80% ($K = 0.05$) (Figures 6A and 6B). To determine whether these rates exceeded chance levels, we generated a null distribution by randomly shuffling module assignments and computing matching rates over 100 iterations. We compared the observed matching rate to the mean of the null distribution and found that the observed matching rate was above the chance level but the difference was less than 0.1% for both within-state and across-state comparisons (Figures 6C and 6D). This small difference suggests that modules may undergo dynamic reconfiguration, although the statistical significance of this difference remains unclear. To address this, we computed a Z score by dividing the difference between the observed matching rate and the mean matching rate of the null distribution by the standard deviation of the null distribution's matching rates. We found that, for both an awake state and sleep, the Z scores of module similarity were consistently greater than 3 (gray dashed line in Figure 6E) across all mice, regardless of whether the comparison was within or between states (Figure 6E). Similarly, for an awake state and anesthesia, within-state comparisons yielded Z scores greater than 3 for all mice, and across-state comparisons exceeded this threshold in three of four mice (Figure 6F). These findings demonstrate that, despite the dynamic reconfiguration of spatially intermixed modules, these neural ensembles exhibit significant temporal stability. Such a temporal stability was also observed across various conditions, including different connection densities (Figure DS1-18), consensus clustering (Figure DS1-19), and different resolution parameters (Figures DS1-20 and DS1-21 for $K = 0.05$), with the exception of $\gamma = 0.5$, where the occasional detection of a single module resulted in chance-level stability (Figures DS1-20C, DS1-20D, DS1-21C, and DS1-21D).

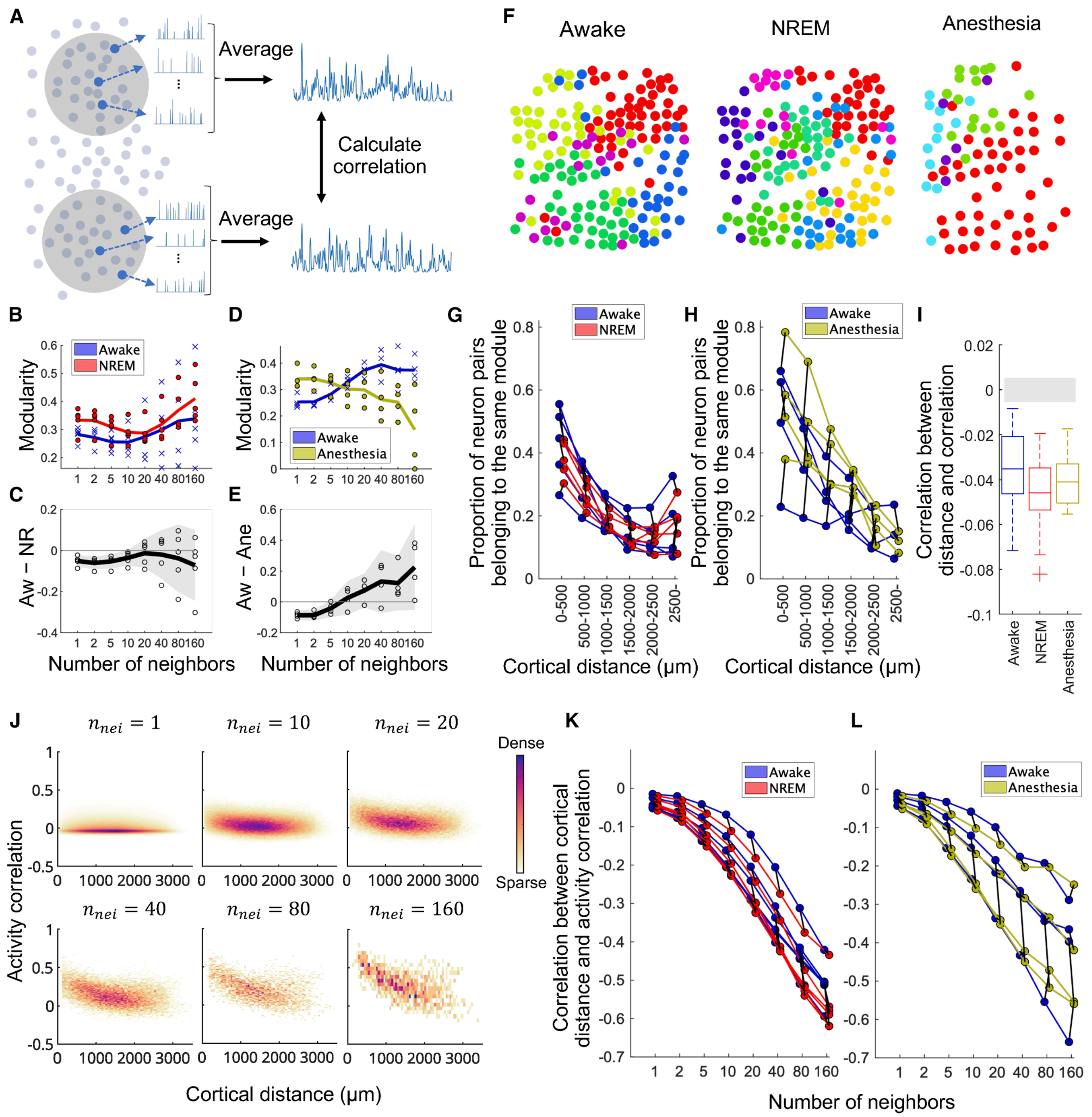

Figure 7. Modularity and distribution of modules at different spatial scales

(A) Schematic of coarse-grained mesoscale functional network estimation.

(B) Comparison of modularity of functional networks between an awake state (blue) and NREM sleep (red) at different spatial scales ($K = 0.05$). Crosses (an awake state) and dots (NREM sleep) represent modularity of individual mice, averaged across networks estimated from different time windows. Lines represent the mean modularity across mice.

(C) Difference in modularity between an awake state and NREM sleep at different spatial scales ($K = 0.05$). Each circle represents an individual mouse. The black line represents the average across mice. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample t test, $n = 5$, $df = 4$).

(D) Same as (B) but comparing an awake state (blue) and anesthesia (yellow).

(E) Same as (C) but comparing an awake state and anesthesia. The gray-shaded area indicates the 95% confidence interval (based on two-tailed one-sample t test, $n = 4$, $df = 3$).

(F) Example of the spatial distribution of coarse-grained functional network modules during each state ($K = 0.05$, $n_{nei} = 40$). Each color represents a specific module.

(legend continued on next page)

Modularity and distribution of modules at different spatial scales

To investigate how modularity and the spatial distribution of modules change depending on spatial scales, we spatially coarse-grained cellular-scale deconvolved Ca^{2+} signals and estimated functional networks (Figure 7A). This is crucial for identifying the spatial scales at which functional networks during sleep and anesthesia exhibit shared changes compared to an awake state. For the spatial coarse-graining, we clustered every n_{nei} neighboring neurons into a single parcel, where n_{nei} ranged from 1 (without coarse-graining) to 160. We then averaged the deconvolved Ca^{2+} signals within each parcel and computed the Pearson correlation between parcels. Subsequently, we binarized the resulting correlation matrices and calculated modularity.

We found that, as the level of coarse-graining increased, network modularity did not show consistent differences across different mice between an awake state and sleep (Figures 7B and 7C) nor between an awake state and anesthesia (Figures 7D and 7E) (see also Figure DS1-22 for changes in modularity values at each spatial scale as a function of K). Specifically, for the comparison between an awake state and sleep, the variability of modularity differences became large, and the 95% confidence intervals included zero when $10 \leq n_{nei}$ (gray-shaded area in Figure 7C). In the comparison between an awake state and anesthesia, the modularity values were, on average, higher during an awake state than anesthesia when n_{nei} was large (Figures 7D and 7E). However, the variability was substantial across mice, and the 95% confidence intervals included zero when $5 \leq n_{nei}$ (gray-shaded area in Figure 7E).

As for the spatial distribution of the modules, we found that the modules were localized during all states in the coarse-grained mesoscale networks (Figure 7F). In other words, unlike cellular-resolution networks (Figure 5), nearby parcels were more likely to belong to the same module than distant ones (Figures 7G and 7H for $K = 0.05$, $n_{nei} = 40$; see Figure DS1-23 for other K values; see Figure DS1-24 for consensus clustering). Here, the distance between parcels was defined as the Euclidean distance between their centroids (i.e., the mean spatial coordinate of their member neurons).

One possible explanation for the change in module distribution with coarse-graining is that correlations between nearby neurons were slightly stronger than those between distant neurons but not strong enough for the module distribution to be localized at the cellular scale. If this were the case, as correlations build up through coarse graining, then the correlations between nearby parcels would become markedly higher than those between distant parcels, leading to module localization.

We investigated how the correlation depended on the distance at each spatial scale and confirmed that this explanation was indeed the case. First, we calculated correlation between cortical distance and activity correlation at the single-cell level and compared it with that of shuffled data. We found a subtle but significant dependence of the correlation on distance during all states (all data points in Figure 7I are below the gray-shaded area) even at the single-cell level. Second, we investigated how the distance dependency of activity correlation changed when we coarse-grained neural activity into different spatial scales. We found that, as the level of coarse-graining increased, the distance dependency of activity correlation became more intense during all states due to the buildup of weak correlations at the cellular level (Figures 7J–7L). This intensified dependence accounts for the module localization in coarse-grained networks.

Complementary analysis of signed and weighted networks

The analyses presented thus far have focused on binarized networks. This approach was chosen because fixing the connection density allows for a consistent comparison of network topology across different brain states, isolating changes in the pattern of connections from the confounding influence of varying edge weights. However, binarization is not the only solid approach, and analyses on signed and weighted networks are also well established.[6]

To confirm the robustness of our results, we performed the complementary analysis on signed and weighted networks and obtained results consistent with those from our primary binary network analysis. Specifically, we confirmed the following key findings. (1) Modularity was higher during sleep and anesthesia than during an awake state across different resolution parameters (Figures DS1-25 and DS1-26; see Figures DS1-27 and DS1-28 for the number of detected modules). (2) Modules were distributed in a spatially intermixed manner (Figures DS1-29 and DS1-30). (3) Modules were temporally more stable than chance levels (Figures DS1-31 and DS1-32). (4) Networks during sleep and anesthesia exhibited greater modularity than during an awake state only at low levels of coarse graining (Figure DS1-33). (5) Nearby parcels of coarse-grained functional networks ($n_{nei} = 40$) were more likely to belong to the same module than distant ones (Figure DS1-34).

Furthermore, we also ensured that our results were not biased by our neuron selection criterion, which included only neurons active across all time windows. We obtained consistent findings even when we constructed networks from different sets of neurons active within each respective time window while keeping the network size constant via subsampling (see STAR Methods

(G and H) Relationships between cortical distance and module assignments of coarse-grained functional networks ($K = 0.05, n_{nei} = 40$) during an awake state (blue), NREM sleep (red), and anesthesia (yellow). The coordinates of each parcel used for distance calculation are defined as the average coordinates of the neurons that constitute the parcel. Each dot represents an individual mouse, with data averaged across networks estimated from different time windows. Data from the same mouse are connected with lines.

(I) Correlation between cortical distance and activity correlation during each state. The boxplots include all windows across all mice. The gray-shaded area represents the range between the upper and lower 2.5 percentiles of the correlation between distance and activity correlation when the position information is shuffled. This percentile range is calculated for each window by generating 1,000 null-model correlations, and the strictest range across all windows is illustrated.

(J) Example of the relationship between cortical distance and activity correlation (during an awake state) at different levels of coarse-graining.

(K and L) Correlation between cortical distance and activity correlation during an awake state (blue), NREM sleep (red), and anesthesia (yellow) at different levels of coarse-graining. Each dot represents an individual mouse, with data averaged across networks estimated from different time windows. Data from the same mouse are connected with lines.

for details). Specifically, modularity was higher during sleep and anesthesia than during an awake state (Figures DS1-35 and DS1-36; see also Figures DS1-37 and DS1-38 for the number of detected modules), and modules were spatially intermixed (Figures DS1-39 and DS1-40).

In addition, we confirmed that, even without the subsampling mentioned above, which means that the number of nodes is different across networks, modularity differences were observed (Figures DS1-41 and DS1-42).

DISCUSSION

In this study, we recorded the spontaneous activities of approximately 10,000 neurons in a contiguous 3 $\times$ 3 mm FOV, including multiple cortices in each mouse (Figure 2), and compared the structure of functional networks between an awake state and sleep, as well as between an awake state and anesthesia, at both the single-cell and mesoscale levels. At the single-cell level, we found that modularity was higher during both sleep and anesthesia than in an awake state (Figures 3C–3F) and that the contribution of the highest-degree neurons to modularity did not necessarily differ between an awake state and sleep or an awake state and anesthesia, while mid- to high-degree neurons contributed more to the modularity difference (Figures 4C–4F). We also found that modules were spatially intermixed during all states (Figure 5) but exhibited a temporal stability (Figure 6). Unlike cellular-resolution functional networks, mesoscale functional networks estimated from coarse-grained neural activity did not show statistically robust differences in modularity between states (Figures 7B–7E), and modules were spatially localized during all states (Figures 7F–7H).

Segregation of functional networks during sleep and anesthesia at the cellular scale

While previous studies using macro/mesoscale recordings have already shown that network segregation increases during both sleep and anesthesia,[9,10,16–19] our findings on the cellular-resolution functional networks represent a conceptually novel contribution rather than an incremental result. This is because the structure of single-cell resolution functional networks cannot be inferred from macro/mesoscale observations alone, while the reverse is possible; i.e., the properties of macro/mesoscale functional networks can be inferred by coarse-graining the single-cell resolution data. Therefore, network segregation observed at the single-cell level provides fundamentally new insights that cannot be extrapolated from macroscale functional networks.

In addition, our result on the decomposition of modularity offers an alternative perspective to brain network studies that focus on hub nodes, those with the highest degrees.[50–52] In general, the highest-degree hubs are thought to play significant roles in the network. Given that an awake and sleep/anesthesia are physiologically and functionally different, one might expect that the highest-degree neurons would have a large effect on the change in modularity depending on the level of consciousness. However, we found that the differences in modularity between an awake state and sleep as well as between an awake state and anesthesia were largely associated with mid- to high-degree rather than the highest-degree neurons. These results suggest that the broader population of neurons contributes to network segregation. It should be noted that, while the contributions of the highest-degree neurons to modularity are similar across states, the identities of these neurons are not necessarily preserved. This is because modules, and thus the degree of neurons, vary across states. It should also be noted that our result does not exclude the possibility that the highest-degree neurons play fundamental roles in brain function. Rather, these hub neurons, whose identity varies across states, may serve fundamental roles in maintaining essential brain functions regardless of consciousness state. These roles could explain their relatively stable and high contribution to modularity across states. Our results highlight the importance of investigating network structure not only through highest-degree hub neurons but also by considering contributions across all neurons. Therefore, our findings suggest that a more inclusive approach to network structure is essential for understanding brain function at a single-cell level. Such an approach is not only relevant to consciousness research but also holds promise for exploring pathological brain conditions from a novel perspective. For instance, while brain networks in individuals with Alzheimer's disease (AD) are known to exhibit disturbed modular structures[57,58] and changes in hub regions,[59–62] future studies could examine how degree-specific contributions to modularity evolve with AD progression. Such investigations could provide a more comprehensive understanding of brain diseases and their relationships to network structures.

Neural mechanisms underlying spatially intermixed modules

Our finding of spatially intermixed modules brings about conceptual advancement by challenging the conventional view on how brain regions interact during unconsciousness.[13] As for spatially intermixing correlation patterns, previous studies utilizing single-cell resolution recording techniques have reported that neurons with different stimulus selectivity are distributed in a spatially intermixed manner within a single modality such as the visual[63] and sensory cortices.[64] The novelty of our finding is that modules are intermixed not only within but also across multi-modal brain regions. Since spatially intermixed modules imply that distant neurons sometimes belong to the same module, our finding suggests that, not only during an awake state but also during sleep and anesthesia, when functional networks are "segregated," long-range functional connections remain across multi-modal brain regions. This is in contrast to conventionally studied macro/mesoscale functional networks, where modules exhibit spatial localization.[17,23,30,56] Such nuanced segregation can only be discovered by simultaneously recording the neural activity during different brain states across multi-modal brain regions at single-cell resolution.

There are at least three possible mechanisms underlying spatially intermixed distribution of modules: common inputs, poly-synaptic inputs, and strong and variable recurrent interactions. First, if there are common inputs to distant neurons, and such common inputs do not depend much on the distance between the neurons, then we would observe spatially intermixed modules. Consistent with this, recent empirical evidence from anterograde tracing methods has shown that the projections of individual neurons are more diverse and divergent than previously indicated.[65] Second, in addition to common inputs,

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poly-synaptic inputs may explain these results. Because of the sampling rate of the recording and the slow kinetics of the Ca^{2+} sensor, we could not distinguish the correlations induced by mono-synaptic direct inputs from those induced by poly-synaptic direct inputs. Although the probability of mono-synaptic connections strongly depends on the distance between neurons, the probability of poly-synaptic connections may not. As correlations in Ca^{2+} imaging are combinations of correlations induced by both mono-synaptic and poly-synaptic inputs, it is possible to empirically observe spatially intermixed modules. Third, if recurrent synaptic connections are strong and variable across neurons, then we would observe salt-and-pepper modules. In fact, recent studies utilizing both optogenetics and computational modeling have shown that salt-and-pepper responses to visual inputs in V1 can be explained by strong recurrent connections that are variable across neurons,[66,67] even though most excitatory connections that a cortical neuron receives originate within a few hundred microns of their cell bodies.[68–70]

To further clarify the underlying neural mechanisms of the module distribution, a direct causality analysis, instead of the correlation analysis employed in this study, is needed. For example, perturbative approaches, such as optogenetic stimulation during imaging, can help assess the existence of causal interactions through direct synaptic connections.[71,72] This step toward elucidating causal relationships may offer deeper insights into the intricate mechanisms that shape spatially intermixed modules.

Bridging the gap between macro and micro

The significance of our multi-scale analysis lies in revealing the spatial scales, from the single-cell scale to the mesoscale, at which changes in functional network structures commonly associated with the reduction of consciousness are observed. We found that the modularity measure could distinguish an awake state from sleep and anesthesia at the single-cell scale but not at the mesoscale (when the number of neighboring neurons $n_{nei} > 10$; Figures 7C and 7E). This finding suggests that investigating functional network structures in relation to the reduction of consciousness requires examining not only the traditionally studied macro/mesoscale but also the single-cell scale. Our study serves as a starting point to inspire future research aimed at identifying the spatial scales that are fundamentally important for consciousness or, more generally, for brain information processing. Such investigations could be pursued experimentally[73–77] or through theoretical modeling and simulations.[32,78–80]

In this context, our study may connect to other coarse-graining methods used in theoretical neuroscience. In particular, recent studies have applied renormalization group theory to neural data to investigate scalings in statistical features, such as the probability of silence, eigenvalues of the covariance matrix, and correlation time.[75,76,81] In contrast, our coarse-graining approach is novel in that it directly investigates the network structure itself across different spatial scales. A promising future direction would be to integrate our multi-scale network analysis with these established theoretical frameworks.

Even when studying mesoscale functional networks, it is important to note that the mesoscale data obtained through direct coarse-graining of single-neuron activity, as in our study, are qualitatively different from those obtained through non-invasive and indirect recording of neural activity, such as fMRI and EEG. In fact, previous studies using macro/mesoscale network analyses based on fMRI and EEG data have reported greater network segregation during sleep and anesthesia compared to an awake state,[8–18] while this result was not observed in our mesoscale network analysis. These discrepancies are likely caused not only by the differences in direct versus indirect mesoscale recordings but also by other factors, including the spatial coverage of the recordings and temporal resolution. For instance, our recordings focused on layer 2/3 neurons, whereas fMRI captures signals from nearly all cortical layers. To further bridge the gap between micro- and mesoscale networks, analysis of single-cell resolution data, as conducted in this study, will be necessary.

That being said, our multi-scale analysis does offer a potential mechanism to reconcile the spatially intermixed patterns we observe with the localized modules reported in macroscale studies.[17,23,30,56,74] Our results suggest that the apparent localization of modules observed in fMRI and EEG data could be an emergent property of spatial averaging. As we demonstrated empirically, even a subtle distance dependency in activity correlations at the single-cell level can be intensified through coarse-graining, leading to the formation of highly localized modules (Figure 7). While we observed this phenomenon in our calcium imaging data, it is plausible that a similar principle holds for other macroscale recordings, where the signal from a single voxel or sensor reflects the averaged activity of a vast underlying neural population.

Regarding the spatial distribution of modules at different spatial scales, our results support two seemingly opposing perspectives of brain network organization, the localistic and holistic views, depending on the activity resolution; i.e., macro/meso vs. micro. The former viewpoint posits that the brain is parceled into localized regions and that each region is responsible for specific information processing. For example, many functional network studies have reported that brain networks are segregated into spatially localized modules[17,23,30,56,74] that are thought to represent groups of nodes that perform similar brain functions and are responsible for specialized information processing.[48] This view is consistent with the theory of brain function localization.[82,83] On the other hand, the latter viewpoint posits that cortical information processing is highly distributed; that is, the brain is not as neatly parceled into localized regions, as shown in the brain atlas. For example, we have previously reported that accurate somatosensory perception,[84] perceptual memory consolidation,[85] and memory enhancement[86] require coordinated activity in the primary somatosensory and secondary motor cortices. It has also been reported that behavior-related signals are represented brain-wide across species,[87] including in mice,[88–90] flies,[91] and worms.[92] In addition, some studies utilizing single-cell resolution recording techniques have reported that neurons with different stimulus selectivity are distributed in a spatially intermixed manner, such as in the visual cortex of rats[63] and in the sensory cortex of mice.[64] These results suggest that the brain functions in a more distributed manner by coordinating different regions as a whole, which is consistent with the holism of the brain function viewpoint.[87,93] Our single-cell level network analysis (Figure 5) supports the holistic perspective, while our coarse-grained network analysis (Figure 7) supports the localistic one. Therefore, our results may provide new insights that can bridge the gap between the two perspectives: brain functions are macro/mesoscopically localized and microscopically intermixed in the cortex.

Limitations of the study

Although our single-cell resolution and wide-field calcium imaging revealed how functional network structure changes with the reduction of consciousness at both micro- and mesoscales, our recordings have several technical limitations that constrain more detailed analyses. First, recording durations were limited; only a few minutes of REM sleep and quiet wakefulness and approximately 20–30 min of wakefulness, NREM sleep, and anesthesia were captured in a single imaging session. This limitation made reliable network analysis during REM sleep and quiet wakefulness difficult and also restricted the investigation of long-term changes in network structure. To obtain sufficient REM sleep episodes and characterize the temporal dynamics of functional networks across brain states, future studies employing extended or continuous imaging or techniques enabling stable long-term recordings will be essential. Second, due to the expression profile of the calcium sensor, our recordings primarily captured excitatory neurons. Inhibitory neurons were largely excluded because simultaneous multi-color imaging was technically challenging with the current microscopy setup. Future work combining multi-color labeling strategies or transgenic approaches will enable simultaneous recording of excitatory and inhibitory populations, providing a more comprehensive understanding of cortical network dynamics across brain states.

We should also note that there are several limitations regarding the interpretations of the results. First, the edges identified in our network analysis represent functional connectivity inferred from correlated activity rather than actual anatomical connections. Verifying these inferred connections will require causal manipulations, such as optogenetic activation of specific neurons and measurement of response probabilities in putative target neurons. Such approaches will be important for validating and refining the functional connectivity inferred from calcium imaging data. Second, caution is warranted when interpreting network measures. Although this study focuses on common changes in network structure shared between NREM sleep and anesthesia, such as increased modularity, decreased global efficiency, and increased transitivity (the latter two measures relate to small-world-ness and small-world propensity investigated in our previous study[45]), such similarity observed in network measures does not imply functional equivalence. In fact, NREM sleep supports functions such as memory consolidation[85] and memory enhancement,[86] whereas anesthesia generally does not. Therefore, future studies investigating diverse brain functions beyond the level of consciousness should not rely solely on the network measures used here but should instead examine these functions from multiple perspectives (e.g., by analyzing temporal dynamics of the network, correlation with behavior, or specific population activity patterns).

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Masanori Murayama (masanori.murayama@riken.jp).

Materials availability

All plasmids and adeno-associated viruses (AAVs) generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

- Awake-sleep and awake-anesthesia data for one mouse is available at Zenodo (https://doi.org/10.5281/zenodo.17667863). Data for other mice will be shared by the lead contact upon request.
- Code for reproducing the main figures is available at GitHub (https://github.com/oizumi-lab/mouse_network_2P).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, D.K., I.O., J.K., M.M., and M.O.; data curation, D.K., I.O., and Y.S.; funding acquisition, D.K., I.O., M.M., and M.O.; data acquisition, I.O., M.K., C.M., K.K., and M.M.; data analysis, D.K., I.O., and J.K.; project administration, M.M. and M.O.; resources, I.O., M.K., C.M., K.K., and M.M.; software, D.K., I.O., J.K., and Y.S.; visualization, D.K., I.O., and J.K.; writing – original draft, D.K., I.O., M.M., and M.O.; writing – review & editing, D.K., I.O., J.K., Y.S., M.K., C.M., K.K., M.M., and M.O.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used ChatGPT, Gemini, and Grok in order to improve language and readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- ○ Global efficiency and transitivity
- ○ Methodological considerations for modularity analysis
- ○ Multi-resolution modularity
- ○ Consensus clustering
- ○ Modularity on signed and weighted networks
- ○ Behavior of participation coefficient and within-module degree Z score
- ○ Analysis with different neuron selection criterion

- QUANTIFICATION AND STATISTICAL ANALYSIS
  - ○ Computation of confidence interval for the difference between an awake state and NREM sleep or between an awake state and anesthesia

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV-DJ-Syn-G-CaMP7.09-WPRE	Ota et al.[45]	N/A
Chemicals, peptides, and recombinant proteins		
Fast Green FCF	Nacalai Tesque, Inc.	15939–54
Isoflurane	Pfizer	N/A
Deposited data		
Mouse two-photon imaging data	This paper	https://doi.org/10.5281/zenodo.17667863
Experimental models: Organisms/strains		
Mouse: C57BL/6JJmsSlc	Japan SLC, Inc.	RRID:MGI:5488963
Recombinant DNA		
pN1-G-CaMP7.09	Shiba et al.[94]	N/A
Software and algorithms		
Falcon	Nikon Corp.	https://www.nikon.com/
MATLAB (versions: 2015b–2023b)	The MathWorks	https://jp.mathworks.com/
Python (version: 3.7.11)	The Python Software Foundation	https://www.python.org/
ImageJ	Schneider et al.[95]	https://imagej.nih.gov/ij/
Clampex (version 10.7)	Molecular Devices	https://support.moleculardevices.com/s/article/Axon-pCLAMP-10-Electrophysiology-Data-Acquisition-Analysis-Software-Download-Page
Low-computational-cost cell detection (LCCD) algorithm	Ito et al.[96]	N/A
Brain Connectivity Toolbox	Rubinov and Sporns[6]	https://sites.google.com/site/bctnet/
Code for network analysis	This paper	https://github.com/oizumi-lab/mouse_network_2P
Other		
Wide-field two-photon microscopy (i.e., FASHIO-2PM)	Ota et al.[45]	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

All animal experiments were performed in accordance with institutional guidelines and approved by the Animal Experiment Committee at RIKEN Animal Care and Use Committee (approval number: W2024-2-013(2)). Wild-type mice (Mus musculus; C57BL/6JJmsSlc; Japan SLC, Shizuoka, Japan) were used. Male and female mice (12–24 weeks old) were used in this study. In all the experiments, the mice were housed in a 12 h-light/12 h-dark light cycle environment with *ad libitum* access to food and water.

The influence of sex on the experimental outcomes was not specifically analyzed and, therefore, represents a limitation of this study.

METHOD DETAILS

Adeno-associated virus (AAV) vector preparation

G-CaMP7.09[94] was subcloned into the synapsin I (SynI)-expressing vector from the pN1-G-CaMP7.09 vector construct. The AAV AAV-DJ-Syn-G-CaMP7.09-WPRE was produced as previously described.[97]

Neonatal AAV injection and surgery

Neonatal AAV injections and surgery for two-photon imaging were performed using previously described protocols[44,45] with slight modifications to the surgical procedure.

Neonatal mice (postnatal day 0–2) were collected from their cages and anesthetized on ice (cryoanesthesia) for 2–3 min before injection, and then mounted in a neonatal mouse head holder (custom-made, NARISHIGE). The AAV (diluted to 4.0×10^{12} GC/mL with 1 $\times$ phosphate buffer saline (PBS), 4 μL per pup) was injected into the neonatal cortex using a glass pipette (Q100-30-15, Sutter Instrument) with a diameter of 45–50 μm. The injection site was located at a depth of 250–300 μm in the frontal area. The injection procedure was completed within 10 min after cryoanesthesia initiation, and the pups were warmed until their body temperature returned to normal. After the pups began to move, they were returned to their mothers.

For surgery, mice (8–20 weeks) were anesthetized with isoflurane (2%). Once the mice failed to react to stimuli, we administered hypodermic injections of the combination agent[98]—Medetomidine/Midazolam/Butorphanol (MMB)—at a dose of 5 mL/kg body weight for the procedure. The MMB solution consisted of medetomidine (0.12 mg/kg body weight), midazolam (0.32 mg/kg body weight), butorphanol (0.4 mg/kg body weight), and saline. After anesthesia with MMB, the head was shaved and placed in head holders (SG-4N, NARISHIGE). During surgery, their body temperatures were maintained at 36°–37°C using a feedback-controlled heat pad (BWT-100, Bio Research Center), and their eyes were coated with an ointment (Neo-Medrol EE Ointment, Pfizer Inc.). After removing the scalp, a 5-mm diameter cranitomy was performed over an area that included the primary somatosensory area of the right hemisphere. The craniotomy was then covered with a 5-mm diameter No.1 cover glass (Matsunami Glass Ind.) and sealed with dental cement (Super Bond, Sun Medical). After cranial window implantation, EEG and EMG were performed as described in previous reports.[85,99,100] Briefly, an EEG screw was implanted over the V2L region in the right hemisphere and over the parietal cortex in the left hemisphere and locked with dental cement. Separate EEG signals were referenced to the screws located in the cerebellar cortex. For the EMG recording, a flexible wire cable was implanted into the neck muscle. A stainless-steel headplate (custom-made, ExPP Co., Ltd.) was then cemented on the skull over the cerebellum as parallel to the window glass as possible. The exposed skull, EEG screws, and EMG wires were covered with dental cement. After surgery, the mice were treated with a pesticidine-reversing agent, atipamezole hydrochloride (ANTISEDAN, Zoetis Inc.) solution at a dose of 0.12 mg/kg body weight. The mice were then placed on a heating pad to recover.

Skull images, including the cranial window of the head-fixed mouse, were acquired from one-photon macroscopic imaging after surgery to estimate the brain regions in the subsequent ROI analysis. The FOV was 13.3 mm $\times$ 13.3 mm and the image resolution was 1,024 $\times$ 1,024 pixels after 2$\times$ spatial binning. The excitation light intensity from a blue LED source (LEX2-B-S, Brain Vision Inc.) and the gain and exposure time of the sCMOS camera (Zyla 5.5, Andor) were tuned so that none of the pixels were saturated.

In vivo two-photon calcium imaging of sleeping and anesthetized mice

In vivo two-photon imaging was performed using a custom-designed wide-field two-photon laser scanning microscope (FASHIO-2PM).[45] To observe the spontaneous sleep-wake status in mice under the microscope, the mice were fixed in a custom-made stage box (ExPP Co., Ltd.) by firmly screwing the headplate to the stage, and then covered with an enclosure to maintain body temperature. The floor of the stage box was covered with the same bedding as that of the home cage. After recovery from surgery, the mice were head-fixed in a stage box daily during the light phase. The duration of head fixes increased daily from 1 min to 2 h in a steady manner to ensure that the head-fixed condition did not stress the mice. More than 2 weeks of habituation allowed the mice to show spontaneous wake/NREM/REM sleep states in the stage box. To observe neural activities during anesthesia, the mouse was anesthetized with 0.6% isoflurane inhalation after 20–60 min-wake recordings. Before each imaging session, the imaging window was cleaned with a cotton swab soaked in acetone. The implanted EEG/EMG electrodes were connected to the connector. Fluorescence was observed in the somas of layer 2/3 neurons (120–150 μm below the surface). G-CaMP7.09 was excited at 920 nm using a tunable Ti:Sa laser (Maitai DeepSee, Spectra Physics) and the laser power was set to 60–80 mW at the front of the objective lens. The fluorescence of G-CaMP7.09 was detected with a GaAsP PMT in the range of 515–565 nm. All imaging sessions were performed for approximately 3–4 h/day during the light phase (zeitgeber time 3–7) for sleep imaging, or during the dark phase (zeitgeber time 15–19) for anesthesia recordings at 7.65 frames/s with 2, 048 $\times$ 2,048 pixels using custom-built software (Falcon, Nikon) and were saved as 16-bit monochrome tiff files.

EEG and EMG recording and analysis

EEG and EMG signals were recorded during the imaging experiments. EEG and EMG signals were amplified and filtered (EEG, 0.1–100 Hz; EMG, 5–300 Hz) using an analog amplifier (MEG-6116, NIHON KOHDEN). The signals were digitized using a 16-bit analog-to-digital converter (Digidata 1440A, Molecular Devices) at a sampling rate of 1 kHz and acquired using Clampex software (v. 10.7, Molecular Devices). To temporally match the EEG/EMG data with two-photon images, the scan-timing signals for two-photon imaging were digitized using the same system. The post hoc sleep state determination was basically performed as described in previous reports.[85,101] EEG signals from the same hemisphere as the imaging were used for post-hoc analysis. Briefly, the root-mean-square (RMS) of EMG signals, EEG delta (1–4 Hz) power (normalized to power in 1–50 Hz), and theta (6–9 Hz) power (normalized to power in 1–4 Hz) were calculated in each 4-s sliding window, which is nearly equal to 30 frames of two-photon imaging. Brain states were classified as follows: an awake state – EMG RMS above a manually defined threshold; NREM – EMG RMS below threshold and delta/theta ratio >*rbin*0.3; REM – EMG RMS below threshold and delta/theta ratio < 0.3. Due to the head-fixed condition, EMG signals were generally lower and less variable, leading to occasional misclassification of quiet wakefulness (immobile wake) as REM. These epochs were manually corrected based on visual inspection. Epochs shorter than 12 s were excluded and merged with adjacent states.

Imaging data analysis

All the obtained images were first subjected to motion correction using NoRMCorre.[102] We then used a low-computational-cost cell-detection (LCCD) algorithm[96] to identify the ROIs corresponding to single neurons. The fluorescence time course $F_{ROI}(t)$ of each ROI was calculated by averaging all pixels within the ROI, and the fluorescence signal of the cell body was estimated as $F(t) = F_{ROI}–r \times F_{neuropil}(t)$, where $r = 0.7$. The neuropil signal was defined as the average fluorescence intensity between 4 and 9 pixels from the boundary of the ROI (excluding all ROIs). The ΔF/F was calculated as $(F(t)–F_0(t))/F_0(t)$, where t is time and $F_0(t)$ is the baseline fluorescence estimated as the 8% percentile value of the fluorescence distribution collected in a ±30 s window around each sample time point.[103] We further corrected the baseline with the part of ‘‘estimate_baseline’’ function of Suite2P.[104] The SNR of each ROI's ΔF/F was calculated using the ‘‘snr’’ function in MATLAB (MathWorks), where the signal was the ΔF/F below 0.5 Hz and the noise was the ΔF/F above 0.5 Hz. To exclude ROIs contaminated by noise, we selected ROIs that satisfied the following conditions: (1) The signal-to-noise ratio of ΔF/F was in the top 90%, (2) ROIs not obscured by nearby blood vessels, (3) the range of ΔF/F values fell between −1 and 10, (4) ROIs with at least one Ca^{2+} transient during recording, (5) the neuropil fluorescence was lower than that of the soma, (6) less than 45% of the frames had a ΔF/F lower than average, and (7) the correlation between ΔF/F and the square root of the EMG values was less than 0.2. See Tables S1 and S2 for the number of ROIs after excluding contaminated ones. We then deconvolved ΔF/F signals using OASIS.[105] Subsequently, we smoothed the deconvolved signals by applying a Gaussian filter with a length of 1.96 s (15 frames).

All analyses to detect blood vessels were performed semi-automatically using ImageJ software[95] as follows. The median image of the two-photon images for 500 frames was converted into the frequency domain via a 2D Fourier transform. After extracting the low-frequency components of the image, an inverse Fourier transform was performed. Then, the blood vessel was detected by thresholding methods plugged in with ImageJ as function ‘‘AutoThreshold.’’ The algorithm ‘‘MinError’’ was used for this.

Estimation of functional networks

To estimate the functional networks, we calculated the Pearson correlation coefficients of the deconvolved Ca^{2+} signal smoothed with a Gaussian filter between the neuron pairs, assuming that neural data was approximately Gaussian and correlations were approximately linear. Although the deconvolved and smoothed Ca^{2+} signal is not necessarily Gaussian, we also confirmed that similar results were obtained even when we use Spearman rank correlation (Figures DS1-43, DS1-44, and DS1-45). In the main figures, only neurons that were not silent in any of the windows in each session were included in the analysis (see Tables S1 and S2 for the number of neurons used for network analysis). The window length was 196 s (1500 frames) for the wakefulness-sleep data and 379 s (2900 frames) for the wakefulness-anesthesia data. These window lengths are the longest duration feasible for analysis across all individuals. For the wakefulness-sleep data, since the sleep state of a mouse typically does not persist for more than 196 s, we segmented and combined the contiguous sections of equivalent states to achieve a cumulative duration of 196 s.

We excluded time frames that could introduce artifacts or ambiguity. We excluded frames with EMG potentials above a threshold in the previous and following 6.5 s (50 frames) to avoid motion-induced artifacts. We also excluded frames in which the same state (either an awake state or NREM sleep) did not persist for more than 39 s (300 frames), to avoid ambiguity in determining whether the mouse was awake or asleep. For the anesthesia data, we excluded the first 1000 frames after the start of anesthesia to avoid the possibility that the state is not yet stable.

We binarized the correlation matrices at a threshold, setting it such that the ratio of the number of edges to all possible node pairs was equal to K.[49] For example, if $K = 0.01$, neuron pairs whose absolute correlation values rank in the top 1 percent are considered to have an edge.

Graph metrics

We calculated graph-theoretical measures using the Brain Connectivity Toolbox[6] and custom-made MATLAB codes.

Modularity

Modularity is defined as the number of edges within modules minus the expected number of such edges,[106]

$$Q = \frac{1}{2l}\sum_{i,j \in G}\left[A_{ij} - P_{ij}\right]\delta\left(g_i, g_j\right), \qquad \text{(Equation 3)}$$

where G is the set of all nodes in the network, l is the number of edges in the network, A_{ij} is an element of the adjacency matrix,

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge between } i \text{ and } j \\ 0 & \text{otherwise,} \end{cases} \qquad \text{(Equation 4)}$$

P_{ij} is the expected number of edges between i and j,

$$P_{ij} = \frac{k_i k_j}{2l}, \qquad \text{(Equation 5)}$$

k_i is the degree of a node i, g_i is the module to which node i belongs, and $\delta(r, s)$ is defined as

$$\delta(r,s) = \begin{cases} 1 & \text{if } r = s \\ 0 & \text{otherwise.} \end{cases} \quad \text{(Equation 6)}$$

We identified a module assignment using the Louvain algorithm,[107] which divides a network into modules to maximize Q, and then calculated the modularity. Because the Louvain algorithm is stochastic, we executed it 200 times for each network at each connection density and selected the partition with the highest Q. In the main figures, we report these max-Q partitions. As for the analyses on distribution and temporal stability of modules, we also report results with consensus partitions[55] in the Supplementary Figures (see "Methodological considerations for modularity analysis" for methodological details).

In addition to this standard modularity analysis, which allows the number of modules to vary across networks, we conducted a supplementary analysis to demonstrate that differences in modularity across states are not driven by variations in the number of modules, as shown in Figures DS1-2 and DS1-3. For this analysis, we standardized the number of modules to the smallest number identified by the Louvain algorithm among the networks within a session, iteratively merging pairs of modules by selecting the pair that minimized the reduction in modularity at each step until the target number was reached.

We introduce a node-specific modularity measure, Q_i, which quantifies the extent to which a node i contributes to the modularity Q,

$$\begin{aligned} Q_i &= \frac{1}{2l}\sum_{j \in G}\left(A_{ij} - \frac{k_i k_j}{2l}\right)\delta(g_i, g_j) \\ &= \frac{1}{2l}\left(k_i^{\text{within}} - \frac{\left(k_i^{\text{within}} + k_i^{\text{between}}\right)\sum_{j \in G_i} k_j}{2l}\right) \\ &= \frac{1}{2l}\left[\left(1 - \frac{\sum_{j \in G_i} k_j}{2l}\right)k_i^{\text{within}} - \frac{\sum_{j \in G_i} k_j}{2l}k_i^{\text{between}}\right] \\ &= \frac{1}{2l}\left[\frac{\sum_{j \notin G_i} k_j}{2l}k_i^{\text{within}} - \frac{\sum_{j \in G_i} k_j}{2l}k_i^{\text{between}}\right], \end{aligned} \quad \text{(Equation 7)}$$

where G_i is the set of nodes that belong to the same module as node i, k_i^{within} is the within-module degree of a node i, and k_i^{between} is the across-module degree of a node i. The last line in Equation 7 shows that Q_i is large when node i has many within-module connections, i.e., k_i^{within} is large, and few across-module connections, i.e., k_i^{between} is small. Comparing Equation 3 and the first line in Equation 7, we can see that Q_i is the exact decomposition of the modularity index Q,

$$Q = \sum_{i \in G} Q_i. \quad \text{(Equation 8)}$$

Although participation coefficient and within-module degree Z score are the most commonly used measures for identifying connector nodes and provincial nodes,[108,109] these measures are not directly related to the modularity of the network. On the other hand, Q_i is the exact decomposition of the modularity index Q, thus enabling us to precisely characterize the specific contributions of individual neurons to the overall modular structure of the network. To clarify the conceptual differences between these conventional measures and Q_i, in "Behavior of participation coefficient and within-module degree z-score", we demonstrate the behavior of participation coefficient and within-module degree Z score and compare them with the Q_i measure.

Global efficiency and transitivity

We also calculated the global efficiency and the transitivity as measures of network integration and segregation. The global efficiency is defined as the average inverse shortest path length,

$$E = \frac{1}{n}\sum_{i \in N} E_i = \frac{1}{n}\sum_{i \in N}\frac{\sum_{j \in N, j \neq i} d_{ij}^{-1}}{n - 1}, \quad \text{(Equation 9)}$$

where N is the set of all nodes in the network, n is the number of nodes, and d_{ij} is the shortest path length between node i and j.[110] This measure quantifies how efficiently one node can communicate with another. The transitivity is defined as the fraction of triangles in a network,

$$T = \frac{\sum_{i \in N} 2t_i}{\sum_{i \in N} k_i(k_i - 1)}, \quad \text{(Equation 10)}$$

where t_i is the number of triangles around node i and k_i is the degree of node i.[111] A large number of triangles implies segregation.

Methodological considerations for modularity analysis

To ensure the robustness of our findings, we performed complementary analyses designed to address two well-documented limitations of modularity maximization algorithms. These are (i) the resolution limit,[112] a property whereby the algorithm may fail to identify communities smaller than a certain scale, and (ii) the degeneracy problem,[53] where numerous distinct partitions can yield near-maximal Q values, making reliance on a single max-Q solution potentially misleading. We addressed these issues by implementing multi-resolution analysis[113] and consensus clustering,[54,55] respectively. The specific procedures are detailed in the following.

Multi-resolution modularity

We recalculated modularity with multiple resolution parameters γ,[113]

$$Q = \frac{1}{2l}\sum_{i,j\in G}\left[A_{ij} - \gamma P_{ij}\right]\delta\left(g_i, g_j\right). \quad \text{(Equation 11)}$$

When $\gamma < 1$, fewer and larger modules are detected; when $\gamma > 1$, more and smaller modules are detected. Intuitively, this reflects the balance between two forces: the first term, $\sum_{i,j\in G} A_{ij}\delta\,(g_i, g_j)$, which favors fewer modules, and the second term, $-\sum_{i,j\in G}\gamma P_{ij}\delta\,(g_i, g_j)$, which favors more modules. Adjusting γ shifts this balance and thus the number of detected modules. We ran the Louvain algorithm 200 times for each resolution parameter (γ = 0.5, 1.5, 2.0) when K = 0.05.

Consensus clustering

When examining module distributions, we not only considered max-Q partitions but also consensus partitions. To perform consensus clustering, for each network at each connection density or each resolution parameter, we constructed from the 200 module assignments an agreement matrix D, where each entry D_{ij} is the fraction of runs in which nodes i and j were assigned to the same module. We then ran the Louvain algorithm once on D and obtained a consensus partition. By default, we used the standard resolution parameter γ = 1.0 for consensus clustering. However, in cases where the original modularity was computed at γ = 0.5, using γ = 1.0 during consensus clustering led to very slow convergence, so we set γ = 0.5 for the consensus step in those instances.

Modularity on signed and weighted networks

To confirm the robustness of our modularity results, we also calculated modularity on signed and weighted networks. The extended modularity index Q for such networks is defined as[114]

$$Q = \frac{1}{v^+}\sum_{i,j\in G}\left(W_{ij}^+ - \gamma P_{ij}^+\right)\delta\left(g_i, g_j\right) - \frac{1}{v^+ + v^-}\sum_{i,j\in G}\left(W_{ij}^- - \gamma P_{ij}^-\right)\delta\left(g_i, g_j\right), \quad \text{(Equation 12)}$$

where $W_{ij}^+ \in \left(0, 1\right]$, $W_{ij}^- = 0$ if the connection between nodes i and j is positively weighted, $W_{ij}^- \in \left(0, 1\right]$, $W_{ij}^+ = 0$ if it is negatively weighted, $v^{\pm} = \sum_{i,j} W_{ij}^{\pm}$ is the sum of all positive and negative connection weights, $P_{ij}^{\pm} = \frac{s_i^{\pm} s_j^{\pm}}{v^{\pm}}$ is the expected connection weight, $s_i^{\pm} = \sum_{j\in G} W_{ij}^{\pm}$ is the strength of node i. In this study, we defined our edge weights $W_{ij}^{\pm}$ directly from the pairwise Pearson correlation r_{ij} between neurons:

$$W_{ij}^+ = \max\left(r_{ij}, 0\right), W_{ij}^- = -\min\left(r_{ij}, 0\right). \quad \text{(Equation 13)}$$

As in the case of binary networks, we ran the Louvain algorithm 200 times for each resolution parameter (γ = 0.5, 1.5, 2.0).

Behavior of participation coefficient and within-module degree *Z* score

Here we show the definition and the behavior of participation coefficient and within-module degree Z score, which are commonly used for identifying connector and provincial nodes, and compare them with the Q_i measure.

Participation coefficient of node i is defined as[108,109]

$$y_i = 1 - \sum_{m\in M}\left(\frac{k_i(m)}{k_i}\right)^2, \quad \text{(Equation 14)}$$

where M is the set of modules and $k_i(m)$ is the number of edges between i and all nodes in module m. It quantifies the diversity of a node's connections across the modules. Typically, a node with high participation coefficient is regarded as a connector hub, which facilitates inter-modular integration.[6]

However, a node with more inter-module but less within-module connections, which contributes more to inter-module integration, does not necessarily have larger participation coefficient. This is because this measure does not quantify the strength of inter-module connections but how evenly these connections are distributed across different modules. Suppose there are three modules, A, B, and C, and consider the participation coefficient of node i, which belongs to module A under two scenarios, Case 1 and Case 2. We denote k_i^{within} as the within-module degree of a node i, k_i^{withB} as the number of connections between node i and nodes in module B, and k_i^{withC} as the number of connections between node i and nodes in module C. In Case 1, $k_i^{\text{within}} = 80$, $k_i^{withB} = 10$, and $k_i^{withC} = 10$. In Case 2, $k_i^{\text{within}} = 10$, $k_i^{withB} = 80$, and $k_i^{withC} = 10$ (Figure DS1-46). Node i has the same participation coefficient in both cases, which is given by

$$1 - \left(\frac{10}{100}\right)^2 - \left(\frac{80}{100}\right)^2 - \left(\frac{10}{100}\right)^2 = 0.34. \quad \text{(Equation 15)}$$

On the other hand, Q_i captures the difference in contribution to modularity between two cases. Node i's contribution to modularity in Case 1, Q_i^{c1}, is given by

$$\begin{aligned} Q_i^{c1} &= \frac{1}{2l}\left[\frac{\sum_{j \notin G_i} k_j}{2l} k_i^{\text{within}} - \frac{\sum_{j \in G_i} k_j}{2l} k_i^{\text{between}}\right] \\ &= \frac{1}{2l}\left[\frac{\sum_{j \notin G_i} k_j}{2l} k_i^{\text{within}} - \frac{k_i + \sum_{j \in G_i, j \neq i} k_j}{2l} k_i^{\text{between}}\right] \\ &= \frac{1}{2l}\left[80\frac{\sum_{j \notin G_i} k_j}{2l} - 20\frac{100 + \sum_{j \in G_i, j \neq i} k_j}{2l}\right], \end{aligned} \quad \text{(Equation 16)}$$

while that in Case 2, Q_i^{c2}, is

$$Q_i^{c2} = \frac{1}{2l}\left[10\frac{\sum_{j \notin G_i} k_j}{2l} - 90\frac{100 + \sum_{j \in G_i, j \neq i} k_j}{2l}\right]. \quad \text{(Equation 17)}$$

If the degree of all nodes is the same in both cases, it follows that the increase in inter-module connections and the decrease in within-module connections in Case 2 from Case 1 contribute to a decrease in Q_i, as

$$Q_i^{c2} - Q_i^{c1} = \frac{1}{2l}\left[-70\frac{\sum_{j \notin G_i} k_j}{2l} - 70\frac{100 + \sum_{j \in G_i, j \neq i} k_j}{2l}\right] < 0. \quad \text{(Equation 18)}$$

Within-module degree Z score of node i is defined as[108,109]

$$z_i = \frac{k_i^{\text{within}} - \overline{k}(g_i)}{\sigma^{k(g_i)}}, \quad \text{(Equation 19)}$$

where g_i is the module containing node i, and $\overline{k}(g_i)$ and $\sigma^{k(g_i)}$ are the mean and the standard deviation of the within-module g_i degree distribution. Typically, a node with high within-module degree Z score but with low participation coefficient is regarded as a provincial hub, which facilitates modular segregation.

However, the within-module degree Z score is sensitive to the standard deviation of the within-module degree distribution of its module, which is not directly related to a node's contribution to modularity. For example, consider a node i within a module g_i. It is possible to vary the distribution of within-module degrees of other nodes in the module (i.e., k_j^{within} for $j \in G_i, j \neq i$) in such a way that the distribution's standard deviation $\left(\sigma^{k(g_i)}\right)$ changes, but its mean $(\overline{k}(g_i))$ does not. This variation also leaves the parameters relevant to Q_i unchanged (such as $\sum_{j \notin G_i} k_j$, $\sum_{j \in G_i} k_j$, and $2L$). Therefore, in such a case, within-module degree Z score z_i of node i will be different, even though Q_i remains the same.

Analysis with different neuron selection criterion

In the main text, our network analysis was performed on a consistent set of neurons that were active across all time windows within a given session. This primary approach was chosen for three key reasons: (i) to ensure a constant network size for fair comparison across different brain states, (ii) to isolate changes in connection patterns from trivial effects of varying neuron numbers, and (iii) to enable the tracking of individual neurons' module assignments across states. However, this selection procedure may introduce a bias by excluding neurons that are active in some time windows but not in others.

To confirm that the neuron selection criterion used in the main text did not introduce a bias into our results, we performed a complementary control analysis with an alternative criterion: selecting neurons that were active within each individual time window. While including all recorded neurons in every window might seem like a straightforward alternative, non-active neurons would simply become isolated nodes that do not contribute to modularity. Therefore, a more informative approach is to construct a network for each window using only the neurons that were active within that window. However, the number of active neurons varies from window to window. To fairly compare the network properties across different windows, we randomly subsampled neurons in each window so that the network size remained constant. The subsample size for a given session was set to the minimum number of active neurons observed across all of its windows. This approach maximized the number of neurons included while maintaining a constant network size throughout the analysis. Functional networks were then constructed from these subsampled populations for each window. The results of this analysis are presented in Figures DS1-35, DS1-36, DS1-37, DS1-38, DS1-39, and DS1-40.

QUANTIFICATION AND STATISTICAL ANALYSIS

Computation of confidence interval for the difference between an awake state and NREM sleep or between an awake state and anesthesia

To assess the statistically likely ranges of the differences in network measures between states for each connection density K, confidence intervals were calculated for the difference. As the network measures were estimated multiple times using different time windows for each state within each mouse, the network measures were first averaged for each mouse and state. The average network measures of sleep or anesthesia were subtracted from those of an awake state for each mouse. Subsequently, the 95% confidence interval of the difference in network measures between an awake state and NREM sleep or between an awake state and anesthesia was computed based on two-tailed one-sample t test. For the wakefulness-sleep data, sample sizes were 5 with 4 degrees of freedom. For the wakefulness-anesthesia data, sample sizes were 4 with 3 degrees of freedom.